

algorithms continuously examine these data streams to identify patterns and anomalies that may indicate emerging faults.

A notable example of predictive maintenance implementation is the collaboration between Southwest Airlines and AIXI, an AI startup specialising in aircraft system diagnostics. Through this partnership, Southwest has equipped its fleet with AI-driven analytics tools capable of real-time monitoring of critical systems. Data from engines, landing gear, braking systems, and other components are continuously transmitted and analysed. When anomalies are detected, such as unexpected turbine vibrations or reduced hydraulic pressure, the system flags the issue and recommends targeted maintenance actions. As a result, Southwest has significantly reduced unplanned maintenance-related delays, improving both customer satisfaction and operational efficiency.

Another notable deployment can be seen at Air France Industries, which has adopted autonomous drones developed by the French technology company Donecle. These drones, equipped with AI and computer vision capabilities, perform detailed visual inspections of aircraft exteriors. Traditionally, such inspections required several hours and involved technicians



Donecle's drone (Courtesy: Global Business Press)

manually examining the fuselage, wings, and tail using scaffolding or lifts. Donecle's drones can now scan an entire narrow-body aircraft in under an hour. Using high-resolution imaging and advanced deep learning algorithms, they accurately detect surface anomalies such as lightning strikes, paint degradation, and structural cracks.

Flight operations optimisation

Artificial Intelligence is reshaping how airlines manage flight operations by enabling more informed, data-driven decision-making. By incorporating factors such as current weather conditions, air traffic patterns, aircraft performance metrics, and fuel consumption data, AI-powered systems help airlines design more efficient flight routes,

anticipate disruptions, and respond quickly to changing conditions. This results in smoother operations, improved punctuality, reduced costs, and environmental benefits.

Historically, flight planning relied heavily on historical data and human judgement, limiting responsiveness to dynamic variables such as sudden weather changes, congested airspace, or airport delays. With AI-driven tools, airlines can now predict these variables in real time and adjust operational strategies accordingly, minimising disruptions and improving overall efficiency.

British Airways is a prominent example of this approach, having invested over 7 million pounds in upgrading its technology infrastructure, including advanced AI systems. These systems support functions such as dynamic flight rerouting, real-time gate allocation, and automated schedule adjustments. For example, when delays occur, AI systems can quickly reassign gates or adjust boarding schedules to ease congestion and maintain smooth airport operations. By analysing data from radar systems, weather forecasts, and historical performance records, AI tools also recommend fuel-efficient routes that avoid turbulence and crowded air corridors.

These investments have delivered measurable results. British Airways has achieved an on-time departure rate of 86%, reflecting the impact of



Remote and Virtual Towers (RVTs) (Courtesy: rohde-schwarz.com)